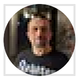
Time Series with Konrad: episode 2

Exponential smoothing: vintage methods



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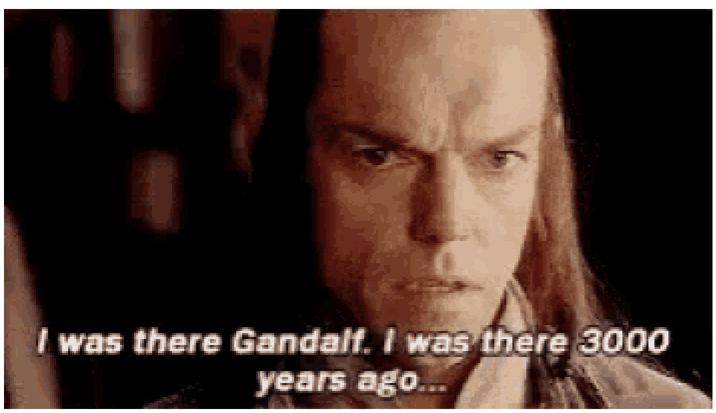
When you think about forecasting the future, you might imagine complex neural networks, sophisticated ML models, maybe an LLM - but some of the most elegant and practical tools emerged in the 1950s. Back in those days, when computers had less memory than your smartphone and practitioners needed fast, reliable short-term predictions that worked with their limited resources.



Pioneered by statisticians like Brown, Holt, and Winters, these exponential smoothing methods revolutionized forecasting with a simple core principle:

recent observations matter more than older ones.

To phrase it more formally: the models create predictions through weighted averages of past observations, where recent data carries the most influence and older ones decay exponentially in importance.

What made them such a big deal was their efficiency: updating a forecast requires only the previous smoothed value and the current observation, which was essential in an era of scarce computing power. They fell out of fashion somewhat as technology advanced, but they're making a comeback lately: high-frequency trading, e-commerce, and real-time monitoring… Anywhere speed and millisecond latency matter, smoothing methods trump marginal gains from more complex algorithms.

In this episode, we'll explore the three most popular methods, while building intuition with real data and discussing their strengths and limitations.

📓 *All code lives in the companion notebook:* https://github.com/tng-konrad/time-series-with-Konrad/blob/main/m02-smoothing-methods.ipynb

Exponential smoothing

Exponential smoothing methods represent a family of forecasting techniques that have been a mainstay of business and statistics since the 1950s. Their enduring popularity stems from their simplicity, computational efficiency, and strong intuitive foundation. At their core, these methods produce forecasts as weighted averages of past observations, with weights that decay exponentially over time. This ensures that recent observations receive greater influence.

Another way to think about exponential smoothing is as a controlled compromise between responsiveness and stability. Recent observations matter more, but past data never fully disappears. This makes the method naturally adaptive to change while still smoothing out short-term noise.

Let's start with the simplest possible assumption: the series has no trend and no seasonality.

Single Exponential Smoothing

Simple Exponential Smoothing (SES) is the foundational method in this family, designed for forecasting time series that exhibit no systematic trend or seasonal pattern. It assumes that the data fluctuates around a stable mean, and the goal is to estimate this underlying level.

Simple Exponential Smoothing (the Brown method) is defined by the relationship:

$$S_t = \alpha X_t + (1-\alpha) S_{t-1} \quad \text{where} \quad \alpha \in (0,1)$$

A few practical observations help make this equation intuitive:

- The smoothed value is a weighted average of the current observation and the previous estimate.
- The smoothing parameter α controls how quickly the model reacts to new data.
- Larger values of α make the model more responsive but noisier; smaller values produce smoother estimates.
- Because the method updates recursively, older observations automatically receive exponentially smaller weights.
- SES is best suited for series without a clear trend or seasonal pattern.

What does the smoothing parameter control?

- **High α**: More weight is placed on the most recent observation, making the model highly responsive to recent changes, but also more sensitive to noise.

- **Low α**: More weight is placed on historical values, producing a smoother forecast that reacts slowly to new information.

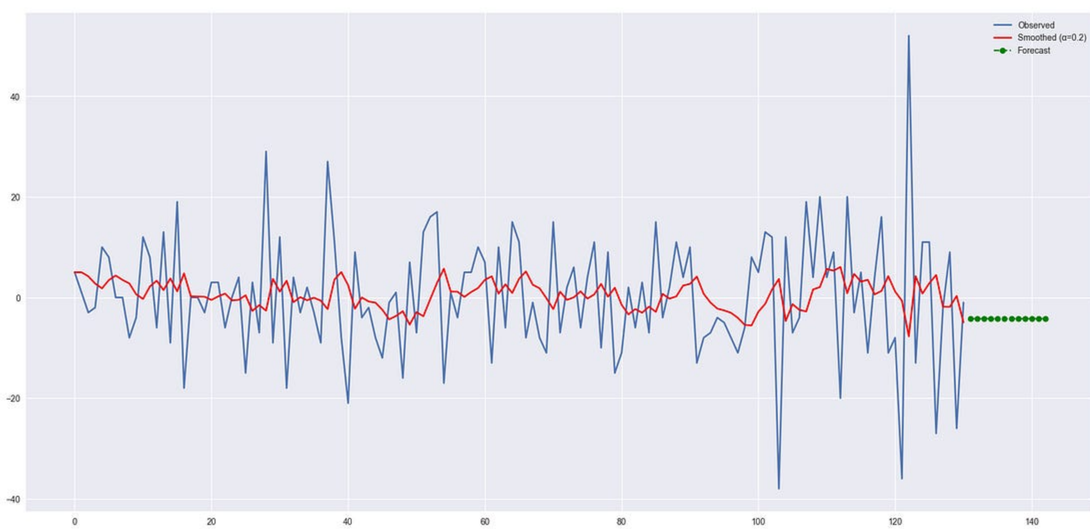
The optimal value of α, along with the initial level, is usually estimated automatically from the data by minimizing forecast errors.

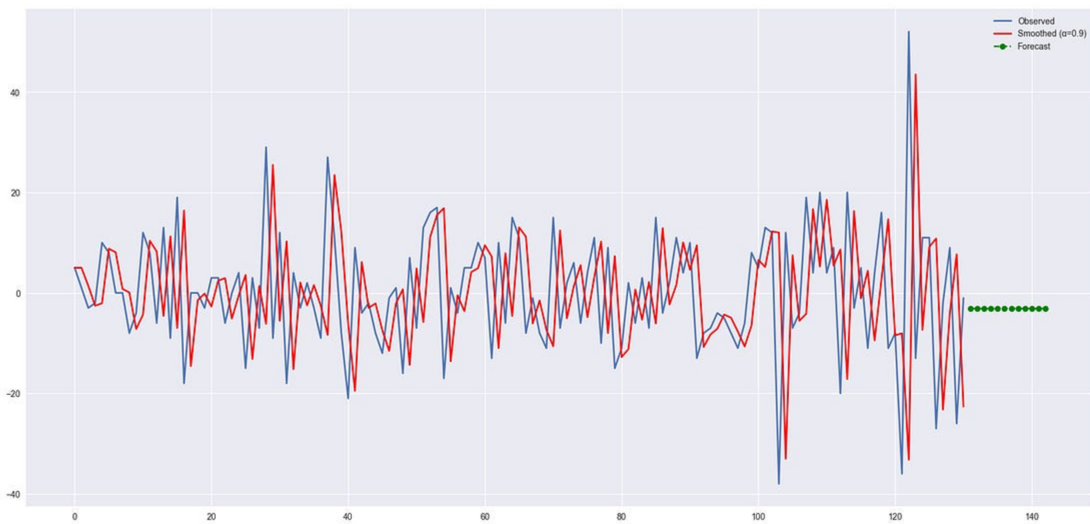
The long-term forecast produced by SES is simple:

$$\hat{X}_{t+h} = S_t$$

Out of sample, our forecast is equal to the most recent value of the smoothed series.

It's an old cliche that a picture is worth a thousand words - so the three pictures below should give you a truly excellent intuition ;-) on how single exponential smoothing works.

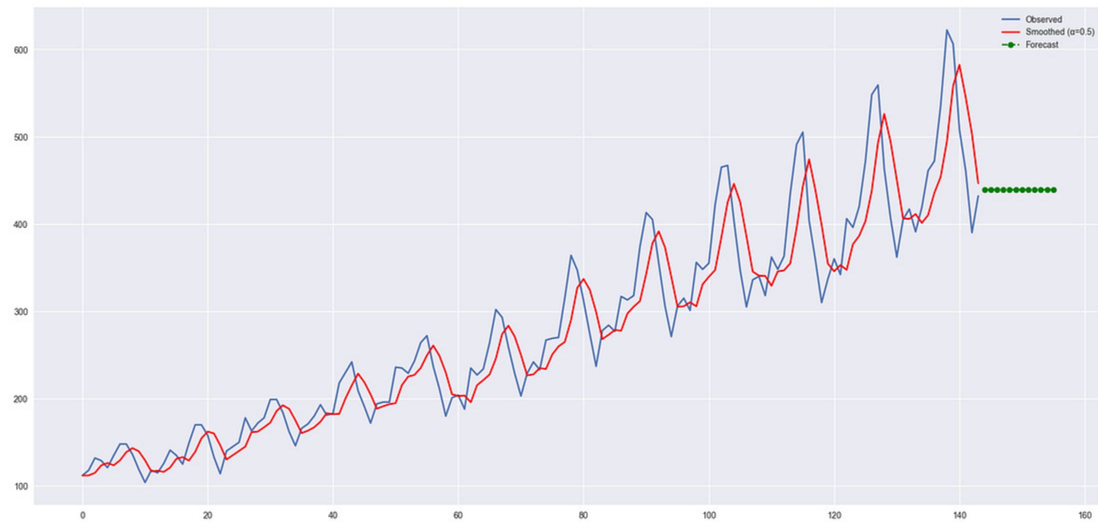




As you can see from the graphs above, for small values of the smoothing constant, most of the variation is removed and the series follows only the general trend. For high values, there is hardly any smoothing at all, and the smoothed series closely tracks the original data (with a delay).

Pro tip: *anytime you are using exponential smoothing that you did not write yourself, double-check the parametrization - does small* **α** *mean heavy smoothing or hardly any at all? The idea that the coefficient closer to 1 means less smoothing is merely a convention.*

What happens if we apply the method to the passengers dataset (introduced in the previous episode)?



On the one hand, the model does exactly what we expect it to do: beyond the range of the original data, it propagates the most recent smoothed value. On the other hand - it is not, to put it mildly, realistic to expect the number of passengers to flatline and the trend to disappear.

This behavior is not a bug, it's a feature: a direct consequence of the model's assumptions. SES assumes the future looks like a noisy version of a constant level. When applied to clearly trending data, it does exactly what it was designed to do and fails in exactly the way we should expect.

Flat forecasts break down as soon as the data starts trending. To fix that, we need to explicitly model the trend.

Double Exponential Smoothing

So far, we have only modeled the level of the series - Holt's method adds one more piece: a smooth estimate of how that level is changing over time.

Simple Exponential Smoothing is inadequate for series that exhibit a trend, as its flat forecast function will consistently lag behind a growing or declining series. Holt's Linear Trend Method, also known as Double Exponential Smoothing, extends SES to explicitly model and forecast data with a trend. This is achieved by adding a second smoothing equation to capture the trend of the series.

We have learned to predict level with our previous methods; now, we will apply the same idea to the trend itself by assuming that the future direction of the time series changes depends on the previous weighted changes.

Double exponential smoothing, a.k.a. the Holt method can be written as:

$$S_t = \alpha X_t + (1 - \alpha)(S_{t-1} + b_{t-1})$$

$$b_t = \beta(S_t - S_{t-1}) + (1 - \beta)b_{t-1}$$

One equation updates the current level by combining today’s observation with yesterday’s level and trend. The second equation updates the trend itself, based on how much the level has changed.

Compared to SES, the key difference is that we now smooth two components: the level and the trend. This introduces a second smoothing parameter, which controls how quickly the estimated trend adapts to new information. As before, these parameters are typically estimated automatically from the data.

The trend smoothing parameter controls how quickly the trend component is updated:

- **High β** gives more weight to the most recent change in the level, which makes the trend estimate highly responsive to recent changes in the series’ growth rate.

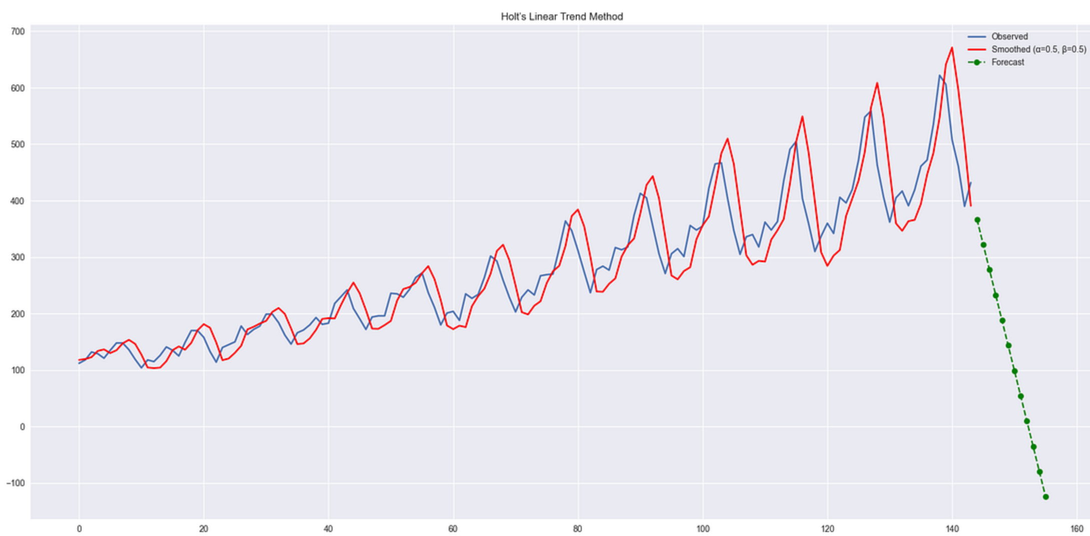
- **Low β** gives more weight to the previous trend estimate, resulting in a more stable trend that changes more slowly over time.

The forecast is defined by

$$\hat{X}_{t+h} = S_t + hb_t$$

The forecast function is no longer flat but trending: *h*-step-ahead forecast is equal to the last estimated level plus *h* times the last estimated trend value.

Let's compare the performance of the two methods on the passenger dataset:



We are moving in the right direction - the forecast going forward is not constant, but follows a trend. However, it is simply an extrapolation of the most recent (smoothed) trend in the data which means we can expect the forecast to turn negative shortly. This is a clear red flag in this domain, and renders the forecast unusable.

Trend alone is still not enough when patterns repeat regularly. This brings us to seasonality.

Triple Exponential Smoothing

If it worked once, maybe it can work twice? A natural extension is to introduce a smoothed seasonal component: Triple Exponential Smoothing, a.k.a. Holt-Winters, is defined by:

$$S_t = \alpha(X_t - c_{t-L}) + (1-\alpha)(S_{t-1} + b_{t-1})$$

$$b_t = \beta(S_t - S_{t-1}) + (1-\beta)b_{t-1}$$

$$c_t = \gamma(X_t - S_{t-1} - b_{t-1}) + (1-\gamma)c_{t-L}$$

One important caveat - always worth repeating: it makes sense to estimate seasonality with period *L* only if your sample size is bigger than *2L*. With fewer than two full cycles, the model cannot reliably distinguish seasonality from noise.

The most important addition is the seasonal component to explain repeated variations around level and trend, and it will be specified by the period. For each observation in the season, there is a separate component; for example, if the length of the season is 7 days (a weekly seasonality), we will have 7 seasonal components, one for each day of the week.

The seasonal smoothing parameter γ controls the evolution of the seasonal pattern over time.

- **High γ** gives more weight to the current period's seasonal index, allowing the seasonal pattern to adapt quickly to changes.
- **Low γ** gives more weight to the historical seasonal pattern, resulting in a more stable seasonal component that changes very little over time.

As we progress from Simple Exponential Smoothing, to Holt, to Holt-Winters, each extension adds flexibility. This allows the model to capture more structure in the data, but it also increases the risk of fitting noise instead of signal. In practice, this means that more complex models are not always better, especially for short or noisy time series. Validation on held-out data is essential.

The forecast *h*-step-ahead is defined as:

$$\hat{X}_{t+h} = S_t + hb_t + c_{(t-L+h)\ mod\ L}$$

Clearly, incorporating both the trend and seasonality explicitly leads to a much higher quality forecast.

Taxonomy

Following *Forecasting: Principles and Practice* by Hyndman and Athanasopoulos, exponential smoothing models can be classified by how they treat trend and seasonality. Each method is labeled using a pair of letters indicating the type of trend and seasonal components.

Some familiar examples:

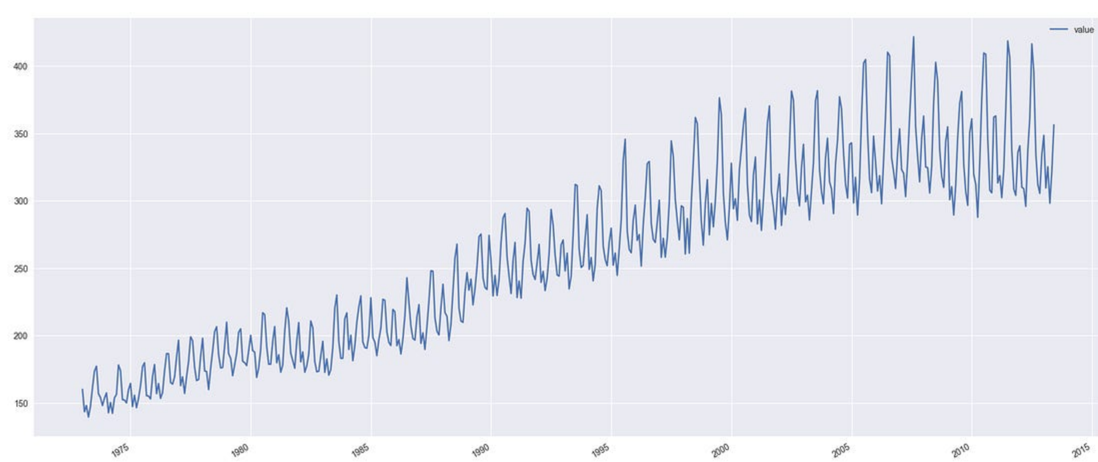
- (N, N): Simple Exponential Smoothing
- (A, N): Holt's Linear Trend method
- (A, A): Additive Holt–Winters

Other combinations are possible, including damped trends and multiplicative seasonality.

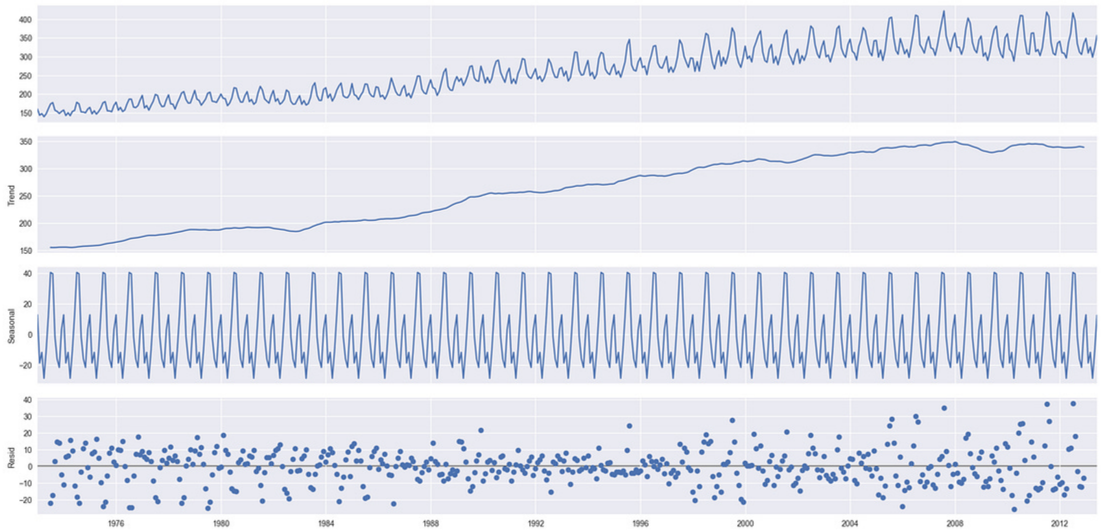
Let's now put these ideas together in a realistic forecasting task. Rather than focusing on a single component, we'll follow a simple workflow: understand the data, choose an appropriate model, validate assumptions, and evaluate forecast accuracy.

Building a model

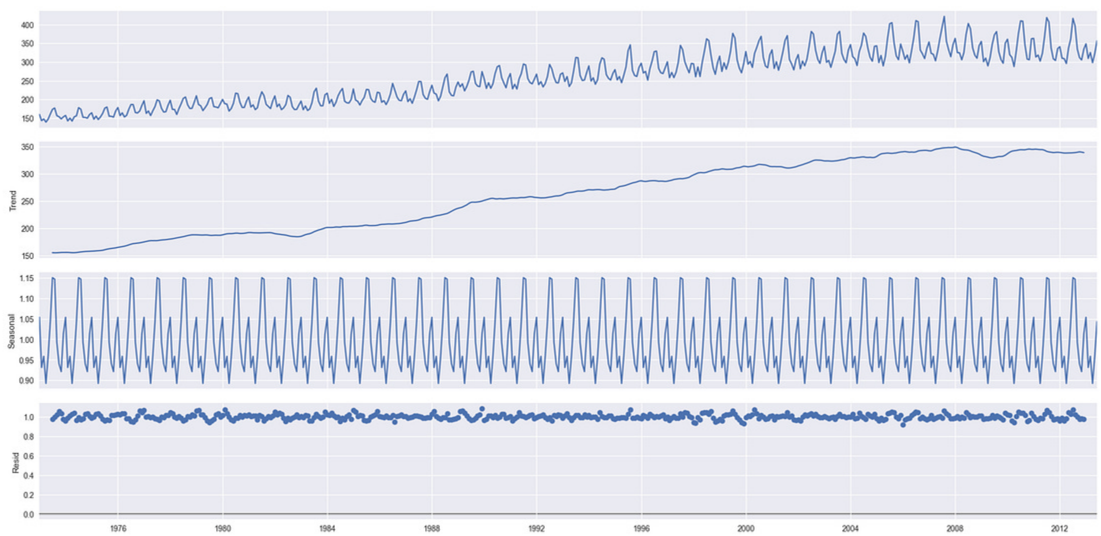
Now that we have all the pieces prepared, we can combine them to build a predictive model. We will use the dataset on daily US energy consumption (in billion kWh)



Let's start with a seasonal decomposition. Additive:



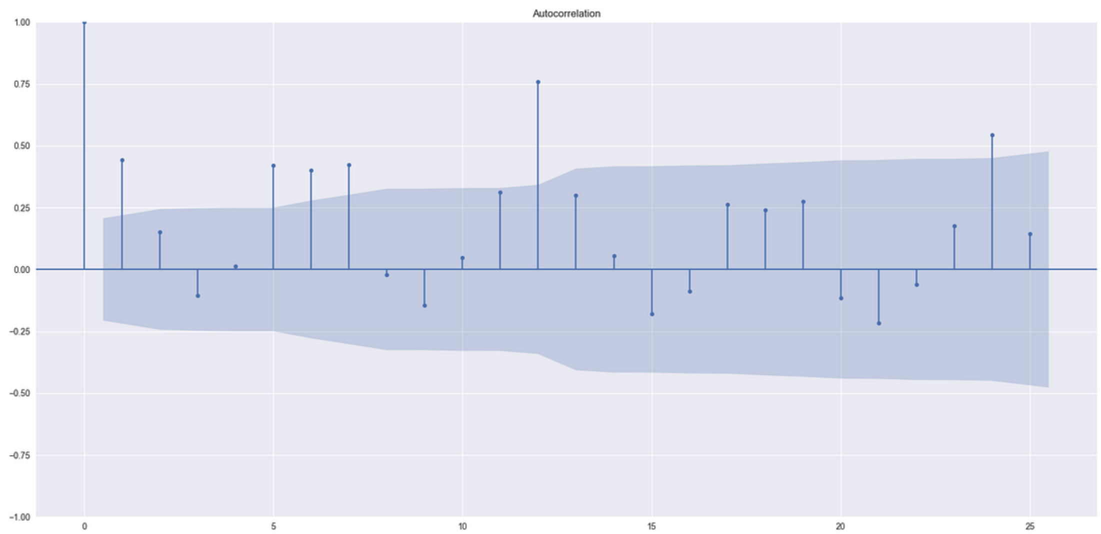
Multiplicative:

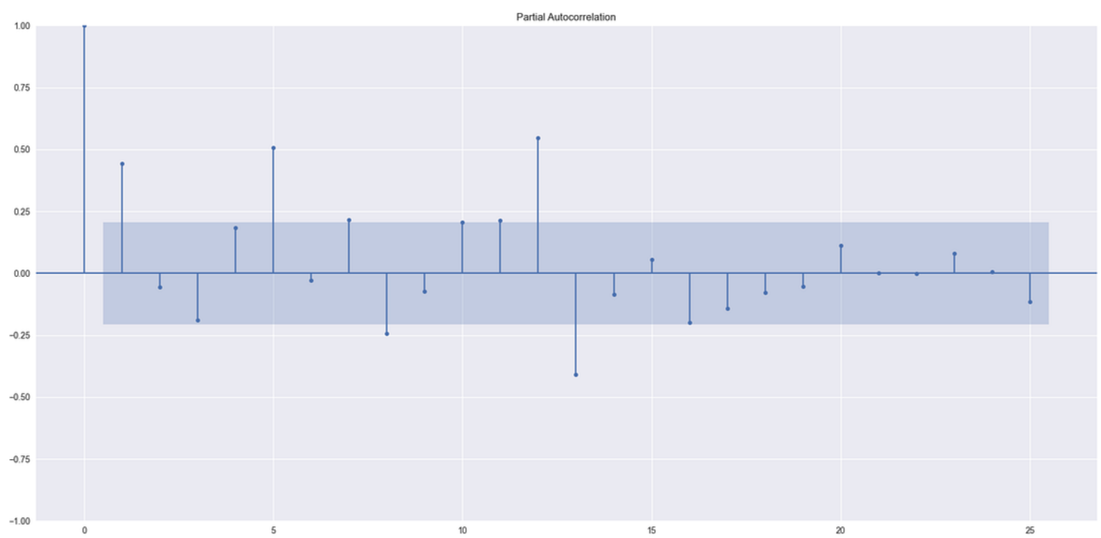


Comparison of the two graphs above shows a few things:

- there is clearly a trend in the data
- we have a clear seasonal pattern
- more stable behavior of the residuals in the second case suggests that a multiplicative decomposition is more appropriate.

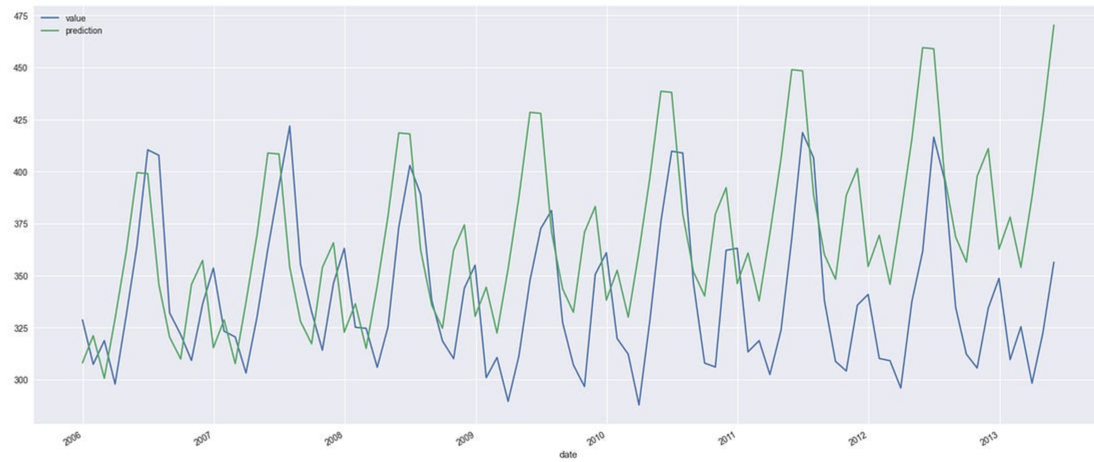
Next, we split the data into training and validation. Our cutoff point will be 2005 - we can fit the model now. What do residuals look like?





The ACF and PACF indicate remaining short-term dependence, suggesting that some dynamics are not fully captured by the smoothing model.

What about the prediction?



The model performs reasonably well for the first few years, but later begins to overestimate consumption. This likely reflects a change in the underlying trend, consistent with the residual diagnostics. While this highlights the model's limitations, its performance is quite acceptable given its simplicity.

The resulting error metrics are:

```
{'MAE': 40.46, 'RMSE': 48.52}
```

These errors are reasonable given the model's simplicity and the long forecast horizon.

Closing time

In this episode, we explored exponential smoothing as a progression of increasingly expressive models: from estimating a stable level, to tracking trends, and finally to capturing recurring seasonal patterns. These methods are simple by design, yet powerful: they are fast, interpretable, and surprisingly effective for short-term forecasting. At the same time, we saw the issues and limitations: smoothing methods extrapolate the recent past and struggle with sudden shifts or changing underlying dynamics in long-term data.

To make the most of exponential smoothing, keep these key principles in mind:

1. **Start simple**: Use Simple Exponential Smoothing for stationary data without trends or seasonality.
2. **Add complexity as needed**: Progress to Double (for trend) or Triple (for trend and seasonality) when your data demands it.
3. **Validate**: Always test on held-out data to ensure your model generalizes.

In the next installment, we'll move beyond smoothing to explore ARIMA models, which offer more flexibility for capturing complex time series patterns by modeling temporal dependence directly.